

Task 2: N-gram Language Models (10 points)

1. What is an N-gram LM?

An **N-gram language model** estimates:

$$P(w_t | w_{t-(N-1)}, \dots, w_{t-1}) P(w_t \mid w_{t-(N-1)}, \dots, w_{t-1})$$

Markov assumption:

The probability of the next word depends only on the previous $N-1$ words, not the entire history.

2. Bigrams and trigrams

Sentence:

“the cat sat on the mat”

Bigrams:

- (the, cat)
- (cat, sat)
- (sat, on)
- (on, the)
- (the, mat)

Trigrams:

- (the, cat, sat)
 - (cat, sat, on)
 - (sat, on, the)
 - (on, the, mat)
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3. MLE for N-grams

For trigrams:

$$P(w_t | w_{t-2}, w_{t-1}) = \frac{\text{count}(w_{t-2}, w_{t-1}, w_t)}{\text{count}(w_{t-2}, w_{t-1})} P(w_t | w_{t-2}, w_{t-1}) = \frac{\text{count}(w_{t-2}, w_{t-1}, w_t)}{\text{count}(w_{t-2}, w_{t-1})}$$

This is **Maximum Likelihood Estimation (MLE)**: probabilities are estimated directly from frequency counts.

4. Data sparsity

N-gram models suffer from data sparsity because:

- The number of possible N-grams grows exponentially with N
 - Many valid sequences never appear in training data
 - This leads to zero probabilities for unseen but reasonable word sequences
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5. Smoothing (example: Add-1 / Laplace)

Add-1 smoothing:

$$P(w | h) = \frac{\text{count}(h, w) + 1}{\text{count}(h) + |V|} \quad P(w | h) = \frac{\text{count}(h, w) + 1}{\text{count}(h) + |V|}$$

Why smoothing is necessary:

- Prevents zero probabilities
- Improves generalization to unseen N-grams
- Makes the model usable in practice